



Ultrasensitive terahertz microfluidic biosensor integrated with tetrahedral DNA nanostructure for specific detection of live cancer cells

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ABSTRACT

Terahertz (THz) microfluidic biosensors have attracted widespread attention in the field of trace biosensing detection due to their non-destructive and harmless properties, but how to effectively achieve specific sensing of living cell samples remains a challenge. Here, a novel microfluidic biosensor consisting of bilayer orthogonal dipole resonant metasurfaces is proposed, which can realize polarization-independent electromagnetic responses and enhance the interaction between biomolecules and structures. The biosensor can achieve high-specific detection for the μL -level trace samples. More importantly, by integrating streptavidin-functionalized tetrahedral DNA nanostructures and biotinylated antibodies as recognition elements on the metasurface, the microfluidic sensor is further endowed with immune recognition properties. This method achieves high-specific trace detection of live lung cancer A549 cells through resonant frequency shifts, creating an extremely low terahertz cells detection limit record down to 1×10^3 cells/mL in the liquid environment. We first prove that tetrahedral DNA nanostructure (TDN) can improve the sensing performance compared to others. Next, the specificity of this method for detecting live lung cancer-like cells is confirmed by the control experiment with ordinary normal cells. Besides, the experiment verified the potential practicality of this method in detecting live tumor cells in complex and multi-biochemical sample environments. This work not only provides a greatly powerful diagnosis tool for live lung cancer cell measurement in the liquid environment but also offers new and reliable insights for the early-stage diagnosis of lung cancers.

1. Introduction

In recent years, terahertz (THz) high sensitivity and real-time metasurface-based biosensors have shown great potential in the detection of cancer cells [1–3]. THz wave refers to the electromagnetic wave with a frequency of 0.1–10 THz ($1 \text{ THz} = 10^{12} \text{ Hz}$) [4]. Due to its non-ionization, non-destructive [5], and its ability to easily capture the vibrational and rotational modes of many biological macromolecules [6], THz wave has become a powerful means to detect protein [7], amino acids [8], nucleic acids [9,10], viruses [11] and cells [12], etc. However, the weak electromagnetic response results in poor sensitivity due to the mismatch between the size of trace biomolecules and the

wavelength of THz [13]. To improve the sensitivity, it is necessary to enhance the interaction between the THz wave and the trace substances.

THz metasurfaces are composed of periodically arranged sub-wavelength structures prepared artificially [14–16], which can generate strong localized fields to enhance the interaction between surface target substances and THz waves to improve sensitivity [17–19]. For example, Nourinovin et al. proposed a THz metasurface based on high Q-factor electromagnetically induced transparency-like resonance for sensitive detection of skin cancer cells [12]. Zhang et al. presented a THz meta-biosensor with high-Q electrical resonance based on interference enhancement of toroidal dipole for sensitive detection of lung cancer cells [20]. However, all the above tests are performed in the dry

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state of cells taking into account the strong absorption characteristics of THz waves by water, which make it difficult to detect living cells in the liquid environments.

Microfluidic chip has the advantages of small size, low cost, and easy and fast analysis, which has attracted extensive attention from researchers. Microfluidic technology can reduce water-induced background absorption by measuring trace samples so that THz waves can capture more useful information from the target analyte [21]. Therefore, the combination of THz metasurface and microfluidic technology is beneficial to enhancing biosensing research of the target substances in the liquid environment. For example, Geng et al. proposed a THz metasurface biosensor integrated with microfluidics for liver cancer biomarker testing [22]. Xu et al. showed a THz microfluidic sensing with toroidal metasurface for enhanced sensitivities of polar liquid analyte [23]. However, all these microfluidic biosensors are based on monolayer-metasurface, where the interaction between the THz wave and the target substances is restricted. Moreover, almost all of the current research on cell-like sensing by THz metasurface biosensors is nonspecific, and the detection limit is relatively high.

Lung cancer is one of the most common malignant tumors in the world given high morbidity and mortality [24,25]. EpCAM, a type I transmembrane glycoprotein, is identified as a tumor cell surface-specific antigen [26]. It has been reported that EpCAM was highly expressed in A549 cells (human lung cancer cell line) [27], and had been explored as the target of antibody therapy [28]. Therefore, it is very important to introduce and modify the EpCAM antibody on the metasurface to make it functionalized and realize the specific capture of A549 cells. Furthermore, considering the inherent stability, nanoscale addressability and versatile functionality of tetrahedral DNA nanostructure (TDN) [29], one can tailor the type of biomolecular probes appended on TDN for the detection of specific biomolecules of interest [30]. In addition, TDN can be used as carriers to firmly anchor

antibodies on the metasurface, and their nano-tetrahedral structure can fully expose antibodies to the liquid environment, which will increase their affinity in the liquid reaction environment. Using this strategy, we can attach EpCAM antibodies to vertices of TDN by existing bio-modification methods, thus achieving specific capture of live cancer cells [31,32].

Herein, we propose a novel microfluidic biosensor with a series of biomodified modified bilayer-metasurface using TDN as carriers, towards the goal of realizing high-specific trace detection of the live A549 cells in the liquid environment. On the one hand, bilayer orthogonal resonance metasurfaces can not only achieve polarization-independent electromagnetic responses, but also can promote more biomolecular-structure interaction events that occur. On the other hand, biotinylated antibodies are connected to TDN on the metasurface through the interaction between streptavidin and biotin, which achieves specific detection of live cancer cells in the liquid environment. Furthermore, we conducted multiple sets of control experiments to fully demonstrate that the proposed sensing method has the capabilities of enhanced sensing and specific detection. Finally, we also demonstrate the potential practicality of the proposed biosensor for detecting tumor cells in complex and multi-biochemical sample physiological environments by detecting live cancer cells in a serum environment. As a proof-of-concept, the experiment results preliminarily demonstrate the feasibility of this THz biosensing method, and are expected to be applied in biochemical detection.

2. Method

2.1. Microfluidic biosensor design

The proposed microfluidic biosensor, as schematic illustration in Fig. 1(a), is composed of the bottom quartz glass substrate of $t_1 = 500 \mu\text{m}$

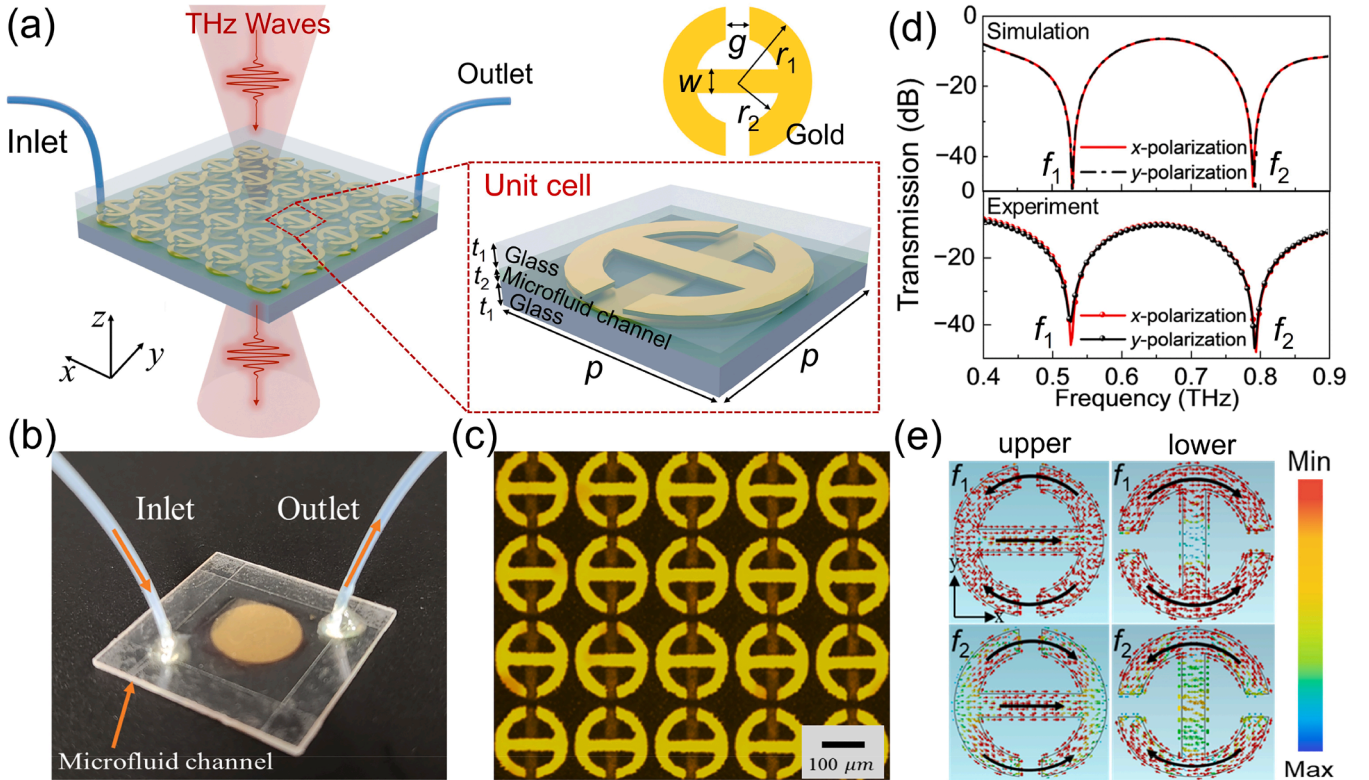


Fig. 1. (a) Art schematic of the proposed microfluidic biosensor and the structural configuration of a unit cell. The structure parameters are listed as: $p = 220 \mu\text{m}$, $t_1 = 500 \mu\text{m}$, $t_2 = 25 \mu\text{m}$, $g = 30 \mu\text{m}$, $w = 30 \mu\text{m}$, $r_1 = 100 \mu\text{m}$, $r_2 = 70 \mu\text{m}$. (b) The physics photo of the proposed microfluidic biosensor. (c) The micrograph of bilayer orthogonal resonant metasurface microstructures in a small zone. (d) Simulated and experimental transmission spectra of the blank microfluidic biosensor for x- and y-polarizations. (e) The upper and lower surface current distributions of metal metasurfaces in the x-y section at resonance peaks f_1 and f_2 , respectively.

thickness, the lower metal metasurface of $t_m = 200$ nm thickness, the microfluidic channel of $t_2 = 25$ μm thickness, the upper metal metasurface of $t_m = 200$ nm thickness and the cap quartz glass of $t_1 = 500$ μm thickness, from bottom to top, respectively. The detailed structural parameters of the metal metasurface are as follows: $p = 220$ μm , $g = 30$ μm , $w = 30$ μm , $r_1 = 100$ μm , $r_2 = 70$ μm . The upper and lower metal metasurfaces are orthogonal arrangements in a periodic manner, which consist of 10 nm thick chromium film and 190 nm thick gold film. Notably, the chromium layer is used for bonding in manufacturing to ensure that the gold microstructures do not fall off. The metal metasurfaces are fabricated through a single ultraviolet laser direct writing system, followed by alignment, curing, packaging, and a series of operations from the proposed microfluidic biosensor, as shown in the diagram in Fig. 1(b) and the micrograph of bilayer orthogonal metasurface microstructures is shown in Fig. 1(c). Different concentrations of trace analytes are injected into the microfluidic biosensor through the inlet and discharged from the outlet. The detailed production process of the microfluidic biosensor chip can be found in Supporting Information S1.

The electromagnetic responses of the blank microfluidic biosensor for different polarization directions are simulated as shown in Fig. 1(d), there are two sharp resonance peaks at 0.53 THz and 0.79 THz, which are marked as f_1 and f_2 , respectively. The theoretical quality factors are $Q_{f1} = 51$ and $Q_{f2} = 82$. The detailed numerical method is in Supporting Information S2. Furthermore, we investigate the experimental transmission spectrum of the blank microfluidic biosensor itself in the x- and y-polarizations by the terahertz time-domain spectroscopy system (THz-

TDS) (see S3 of Supporting Information), as also shown in Fig. 1(d), which is in good agreement with the simulation results. This subtle difference is due to material ohmic loss and manufacturing errors. Besides, the simulated and experimental transmission spectrum shows the polarization-independent of the microfluidic biosensor because the bilayer orthogonal metasurfaces are C4 central symmetric [33]. Fig. 1(e) presents the surface current distributions of the metasurfaces at resonance peaks f_1 and f_2 for the x-polarized plane wave. For the upper metasurface at resonance peak f_1 , it is observed that the toroidal dipolar mode is excited, while the lower metasurface generates the electric dipole resonance mode. At resonance peak f_2 , the upper and lower metasurfaces all exhibit the electric dipole resonance mode, but in opposite directions. The strong interaction between the upper and lower metasurfaces results in two sharp resonance peaks observed. Such structures in our designed configurations are polarization-independent and can effectively enhance the interaction between THz waves and analytes between the bilayer metasurfaces. In addition, the theoretical sensing performance and analysis of related structural parameters for the proposed microfluidic biosensor are provided in Supporting Information S6.

2.2. Synthesis and characterization of tetrahedral DNA nanostructure

Tetrahedral DNA nanostructure (TDN) is developing rapidly and is applied to the biochemical sensing field, and it can be easily synthesized by the principle of base complementary pairing [29,34,35]. The

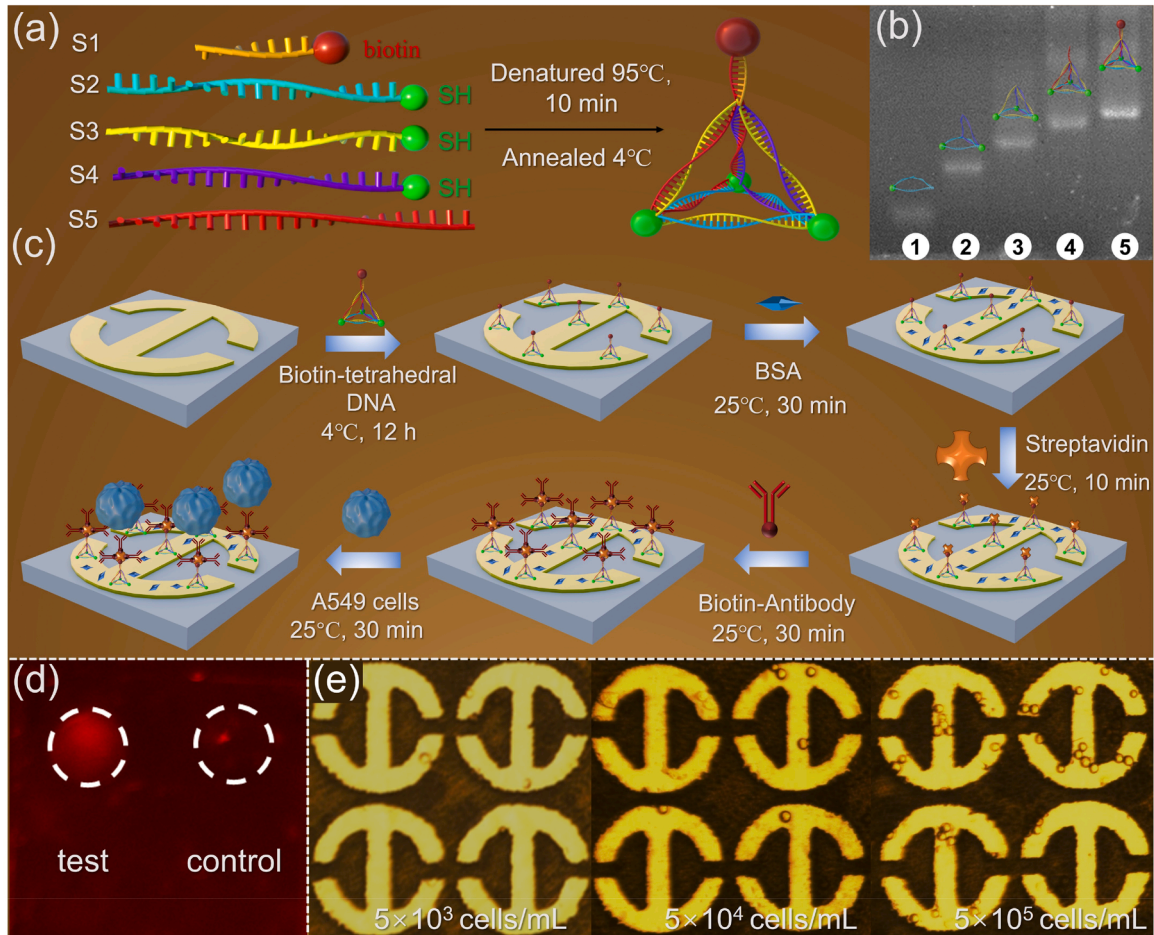


Fig. 2. (a) Self-assembly schematic diagram of tetrahedral DNA nanostructure (TDN). (b) Electrophoresis images for confirmation of the successful synthesis of TDN. Lane 1: S1, lane 2: S1 + S2, lane 3: S1 + S2 + S3, lane 4: S1 + S2 + S3 + S4, lane 5: S1 + S2 + S3 + S4 + S5. (c) The process of the functionalized modified metasurface microstructures. (d) Fluorescence control detection for confirmation of the successful embellishing of TDN. (e) The micrograph of live A549 cells on the functionalized modified metasurface microstructures with different concentrations of 5×10^3 cells/mL, 5×10^4 cells/mL, and 5×10^5 cells/mL, respectively.

self-assembly of TDN is conducted by mixing five single-stranded DNA, followed by denaturation at 95 °C for 10 min and then rapid annealing in a 4 °C [36], the process is shown in Fig. 2(a). All DNAs are synthesized by Sangon Biotech (Shanghai, China), and the related sequences are listed in Supporting Information Table S1. To verify the successful assembly of TDN, the polyacrylamide gel electrophoresis (PAGE) is necessary conducted, as described in Fig. 2(b). Clearly, the TDN (lane 5) moves more slowly than other combinations formed by less than five single-stranded DNA (lanes 1–4), which means the successful preparation of TDN. Subsequently, we carried out characterization analysis using Atomic Force Microscopy (AFM) (see S4 of Supporting Information), confirming the successful synthesis of independent TDN.

2.3. Functionalized modification of the metasurfaces and cells capture

Fig. 2(c) shows the functional modification process of the metasurfaces: 1) Firstly, the microfluidic chip is cleaned with PBS at least three times, then the thirty microliters of 1 μ M biotin-TDN solution is dropped into the microfluidic channel until it is filled and placed for 12 h at 4 °C, followed one remove the unbound TDN by washing with PBS. The self-assembled TDN with site-specifically modified thiol groups can be readily anchored on the surface of the upper and lower gold metasurface microstructures with high reproducibility through the Au-S bonds. To prove that the TDN is successfully modified on the metasurface, we conducted a verification experiment. The modification process for the test and control groups can be found in Supporting Information S5. The chip of the test group is modified with tetrahedron/streptavidin, which is then incubated with the single-stranded DNA modified with Cy5 fluorescent molecules at the 5' end and biotin at the 3' end (see Strand S6 in Supporting Information Table S1). After three washes to remove unbound ssDNA, the chip is imaged under a fluorescence imager. The contradistinctive result is shown in Fig. 2(d), the test group shows red fluorescence in the entire area, while the control group shows no fluorescence indicating the successful modification of TDN. 2) Then we add the thirty microliters of 1 % BSA solution for 30 min at 25 °C, followed by washing off excess BSA by rinsing with PBS. The BSA solution will fill and occupy areas not modified by TDN. 3) Next, thirty microliters of 5 mg/mL streptavidin are added to the microfluidic channel for 10 min at 25 °C to strongly bind to the biotin on TDN, and then one uses the same cleaning way to remove excess streptavidin. The binding mode of streptavidin and biotin is through the highly specific binding between one streptavidin molecule and four biotin molecules. 4) Next, we continue to inject thirty microliters of 0.5 mg/mL biotin-antibodies into the microfluidic channel for 30 min at 25 °C and afterward remove the excess biotin-antibodies. So far, the functionalization of the microfluidic channel has been completed.

Then, we inject different concentrations of live A549 cells cultivated in PBS into the microfluidic channel and incubate at 25 °C for 30 min. Live A549 cells contain EpCAM antigen, and the biotin-antibodies mentioned above are EpCAM-type antibodies, which can specifically capture live A549 cells. Fig. 2(e) describe the visible micrographs of different concentrations of live A549 cells on the functionalized modified metasurface microstructures, which suggest that the live A549 cells can be specifically captured by using the proposed functionalized modified method. Importantly, these captured live A549 cells maintain a round and plump physiological active state with clear cell boundaries, which also proves that this method does not cause damage to the live cells.

3. Results and discussion

In the experiment, A549 cells are chosen to verify the performance of the microfluidic biosensor. The A549 cells are adherent cultured in vitro in Dulbecco's modified Eagle's medium (DMEM) containing 10 % fetal bovine serum (FBS) and 1 % penicillin-streptomycin at 37 °C in a humidified atmosphere (5 % CO₂). The adherent A549 cells can be

suspended in the medium after digestion. The resuspended live A549 cells are diluted with PBS to the indicated concentrations. The 50 μ L live A549 cells are slowly injected into the microfluidic channel and incubated at 25 °C for 30 minutes to allow for a full reaction. Before measurement, the microfluidic biosensor chip is washed three times with PBS to remove cells that have not been specifically captured.

The THz sensing experiments in this work are performed by THz-TDS, aiming to validate the detection capability of the microfluidic biosensor. The terahertz transmission spectra signals can be effectively detected when varying the analytes in the microfluidic channel. In the text, we only discuss the experimental results of the x-polarized considering the polarization-independent characteristics of microfluidic biosensors, and the other is discussed in Supporting Information.

Based on the above functionalized modified microfluidic biosensor, we further investigate the transmission response of live A549 cells at various concentrations at two resonance peaks under x-polarized incidence, as shown in Fig. 3(a) and (b). The transmission spectrum of two resonance peaks all gradually redshifts with the increase of cell concentrations, which proves that the microfluidic biosensor can detect live A549 cells in the liquid environment. Moreover, compared with the transmission reference signal of the microfluidic sensor chip without cells (0 cells/mL), the maximum frequency shifts of the two resonance peaks f_1 and f_2 when the cell concentration reaches 1×10^6 cells/mL are 47.92 GHz and 54.17 GHz, respectively. The experimental results for the y-polarized incidence in Supporting Information S7 show the same trend.

The limit of detection (LoD) is one of the key indicators of biosensors. As shown in Fig. 3(c), the frequency shift of resonance peak f_1 is almost 0 GHz. Fortunately, a redshift of the resonance peak f_2 (2 GHz) can be clearly observed, which means that the microfluidic biosensor can utilize high-frequency resonance peak f_2 to achieve the LoD of 1×10^3 cells/mL. Such a lower detection limit can be attributed to the trapping ability of the bilayer functionalized modified metal metasurfaces to live A549 cells and the enhanced effect of wave-substance interactions. Fig. 3(d) further visually shows the frequency shift histograms of the two resonant peaks at different live A549 cell concentrations. The results show that the frequency shift of low-frequency resonant peak f_1 is slightly smaller than that of high-frequency resonant peak f_2 , which is derived from the theoretical quality factors $Q_{f1} < Q_{f2}$, that is, the local electric field enhancement of resonant peak f_2 is greater, which is also the reason why high-frequency resonant peak f_2 above has a lower LoD. The error bars in Fig. 3(d) are calculated from three repeated experiments, reflecting the reliability of the experiment.

To prove that TDN can improve the sensing performance, we design the control experiments to directly assess the performance of simple double-stranded DNA (dsDNA) modified with terminal thiol and biotin, synthesized from a mixture of S7 and S6 (Table S1), against that of TDN. The experimental measurement spectra of x-polarized are shown in Fig. 4(a) and (b), respectively. Similar results are also observed in the case of y-polarization in Supporting Information S8. As shown in Fig. 4 (c) and (d), multiple experiments have verified that the frequency shift induced by the microfluidic biosensor modified with TDN is larger than that caused by the one modified with dsDNA, meaning that the TDN group possesses a higher sensing performance compared to the dsDNA group. This is mainly due to their different spatial structures.

TDN possesses a three-dimensional structure, high rigidity and desirable biostability [37]. The distribution of antibodies at the TDN vertices is more flexible and diverse, which can capture antigens from multiple directions simultaneously. Besides, TDN can reduce the possibility of entanglement DNA probes [38] and maintain both structural and functional stabilities for a longer duration [39,40]. Those can greatly improve the binding efficiency of antigens. In contrast, dsDNA has a linear structure, the spatial extension and antigen binding mode are relatively limited, which leads to a relatively lower probability and poorer flexibility of dsDNA in adsorbing antigens. Therefore, TDN is preferred in liquid environments and we all adopt the microfluidic

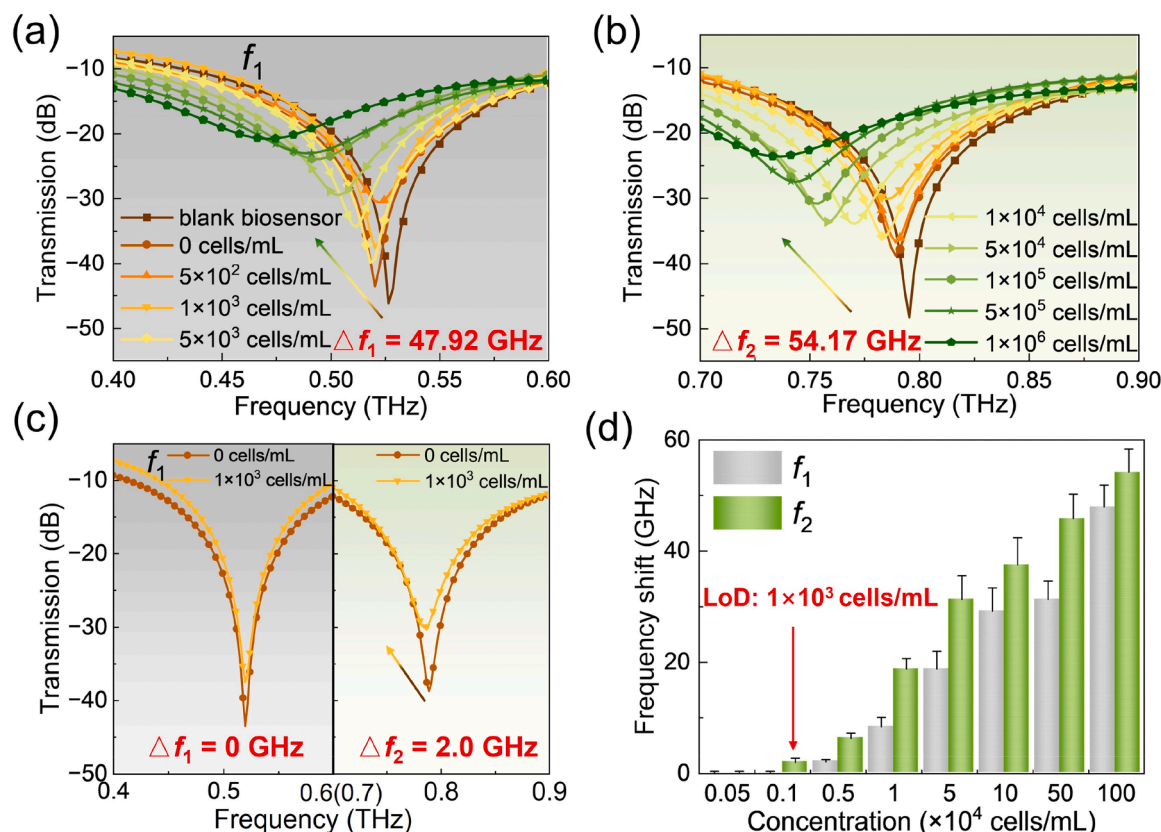


Fig. 3. The experimental transmission spectra of the live A549 cells with different concentrations are obtained by the proposed biosensor at resonance frequencies (a) f_1 and (b) f_2 under the x-polarized. (c) Comparison of detection limits in different concentrations. (d) The frequency shifts variations as the concentration changes at resonance peaks f_1 and f_2 . All experiments are repeated three times.

biosensor modified by TDN to carry out the related works in the subsequent experiments. The error bars and the significant difference (p -value) are calculated from three repeated experiments, reflecting the reliability of the experiment.

To prove the sensing enhanced of antibody (Ab), we conduct an experimentally measured transmission spectrum for the bilayer-metasurface with and without modification of antibody, the experimentally measured results are shown in Fig. 5(a) and Fig. 5(b), respectively. Note that, except for the difference between with and without antibodies, the other modification processes of the microfluidic biosensor remain unchanged according to Fig. 2(c). The resonance peaks in both two cases all are redshifts, but the redshift of with antibody is more obvious. The specific frequency shifts change at different concentrations with and without antibodies are compared in Fig. 5(c) and Fig. 5(d), respectively. It can be found that the frequency shift of the microfluidic biosensor with antibody is more than 45 GHz at the concentration of 5×10^5 cells/mL, compared with less than 15 GHz without antibody, the sensing response improved by more than three times for resonance peak f_2 . However, the detection ability of the antibody-modified microfluidic biosensor is up to 4.53 times that of the unmodified antibody microfluidic biosensor for resonance peak f_1 . More importantly, there are still frequency shifts for the microfluidic biosensor with antibodies even if at the lower concentration of 5×10^3 cells/mL. Here are the experimental results measured under x-polarized, the results of y-polarized show the same trend, which can be found in Supporting Information S9. In short, the sensing capability of the biosensor with antibodies is far stronger than without antibodies.

Next, to verify the specific detection ability of the functionalized microfluidic biosensor, we compare and measure the live A549 cells and the normal cells (normal liver cells L02). The functionalization process of the microfluidic biosensor can be found in Fig. 2(c), where the biotin-

antibody is an EpCAM-antibody, can specifically capture live A549 cells containing EpCAM antigen, but has no recognition effect on normal cells without EpCAM antigen [41,42]. The immune interaction between antigen and antibody makes the microfluidic biosensor chip highly selective. The experimental measured result is shown in Fig. 6(a), there is a significant difference between the live A549 cells and the normal cells at three concentrations. From Fig. 6(b), the live A549 cells acquire a more obvious frequency shift than the normal cells at the same concentration, showing stronger detection ability. Among them, at the highest concentration for resonance peak f_1 , the specific recognition is approximately 7.44 times higher than the non-specific recognition. Therefore, the functionalized microfluidic biosensor can specifically distinguish lung cancer cells and normal cells given the sensitive selective response. For experimental results under y-polarized, as discussed in Supporting Information S10.

The live A549 cells used in the above experiments are all in the ideal PBS environment. Considering the actual situation, this experiment explores the detection of A549 cells in a more complex serum environment to be closer to the actual clinical stage. We use the medium containing 10 % FBS to simulate the physiological environment of tumor cells, and add a certain number of cells into this medium to maximize the simulation of blood samples.

The experimental spectrogram under the x-polarized is shown in Fig. 7(a), the results demonstrated that there are all significant frequency shifts for different concentrations. Subsequently, we plot the corresponding frequency shifts comparison diagram for the PBS and serum environments in Fig. 7(b), respectively. The frequency shifts in the serum environment are slightly larger than those in the PBS environment, which may be due to the interference of various biomolecules in the serum. However, despite the interference of many other components in serum, the microfluidic biosensor can still specifically capture

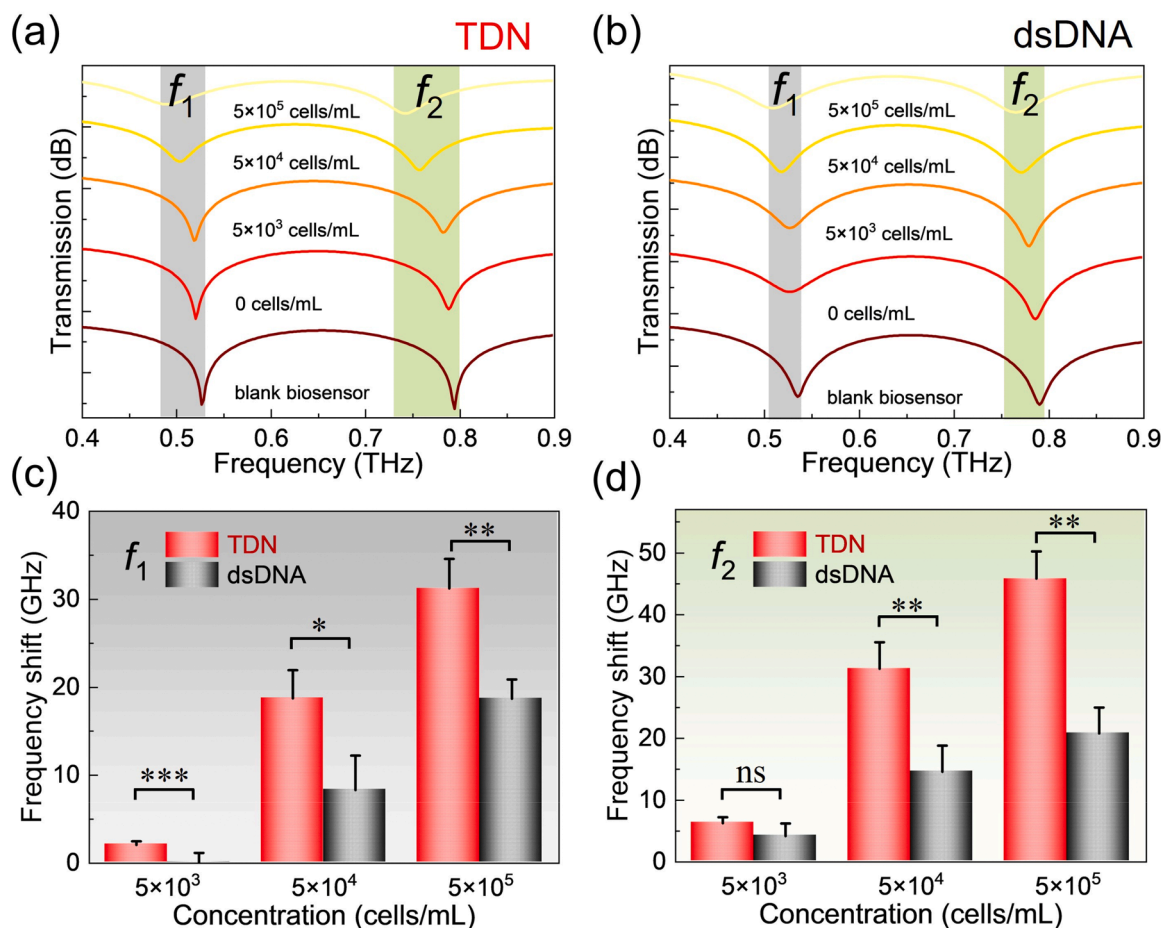


Fig. 4. The experimental x-polarized transmission spectra of the live A549 cells using microfluidic biosensors modified with (a) TDN and (b) dsDNA. Corresponding frequency shifts comparison at resonance peaks (c) f_1 and (d) f_2 . Note: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, ns: no significance. All experiments are repeated three times.

live A549 cells, and most non-target substances will flow out with the cleaning solution before detection, which relatively reduces the interference of residual substances. The experimental results for the y-polarization direction in Supporting Information S11 are still acceptable. Therefore, our microfluidic biosensor holds potential practicability for monitoring tumor cells in complex and multi-biochemical sample environments.

To evaluate the sensing performance of the proposed THz microfluidic biosensor for tumor cells, we compare multiple metrics with those of some representative biosensors from recent years in Table 1. The results show that the proposed microfluidic biosensor can increase the LoD by at least an order of magnitude and can detect trace amounts of live cells with high specificity in the liquid environment, which shows great application potential in the field of THz biochemical sensing.

Although THz technology still has a certain gap in sensitivity compared to electrochemical technology [47] at this stage, it exhibits unique advantages in biochemical detection as an emerging technology. Compared to electrochemical technology, THz technology can achieve non-invasive, non-destructive detection and has high stability. Compared to the fluorescence-based method, THz technology can achieve label-free detection and has signal stability. We believe that THz sensing will gradually mature and be put into application with the in-depth development of technology.

4. Conclusion

In conclusion, a novel microfluidic biosensor consisting of a bilayer-metasurface is proposed for high-specific trace detection of live A549

cells in the liquid environment. The bilayer-metasurface can realize polarization-independent electromagnetic responses and promote a larger number of biomolecule-structure interaction events that occur. By integrating streptavidin-functionalized tetrahedral DNA nanostructures and biotinylated antibodies as recognition elements on the metasurface, the microfluidic sensor is further endowed with immune recognition properties. The experimental results show that the proposed biosensor can achieve highly specific trace detection of live A549 cells in the liquid environment and the detection limit is as low as 1×10^3 cells/mL, which is at least an order of magnitude improvement compared to previous work in the field of THz sensing. Furthermore, the maximum detection ability of the biosensor modified with antibody is 4.53 times higher than that of the antibody-unmodified biosensor. Importantly, the specific response of this biosensor is approximately 7.44 times higher than the non-specific response. In addition, this biosensor has the potential practicability for detecting live tumor cells in complex and multi-biochemical sample physiological environments. The proposed biosensing method shows the detection characteristics of specificity, and low detection limit, which will play an important role in early cancer screening.

CRediT authorship contribution statement

Peng Shen: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Xinmin Yue:** Writing – review & editing, Resources. **Yunyun Ji:** Writing – review & editing, Supervision, Project administration, Formal analysis. **Shengjiang Chang:** Writing – review & editing, Supervision, Resources, Project administration. **Fei**

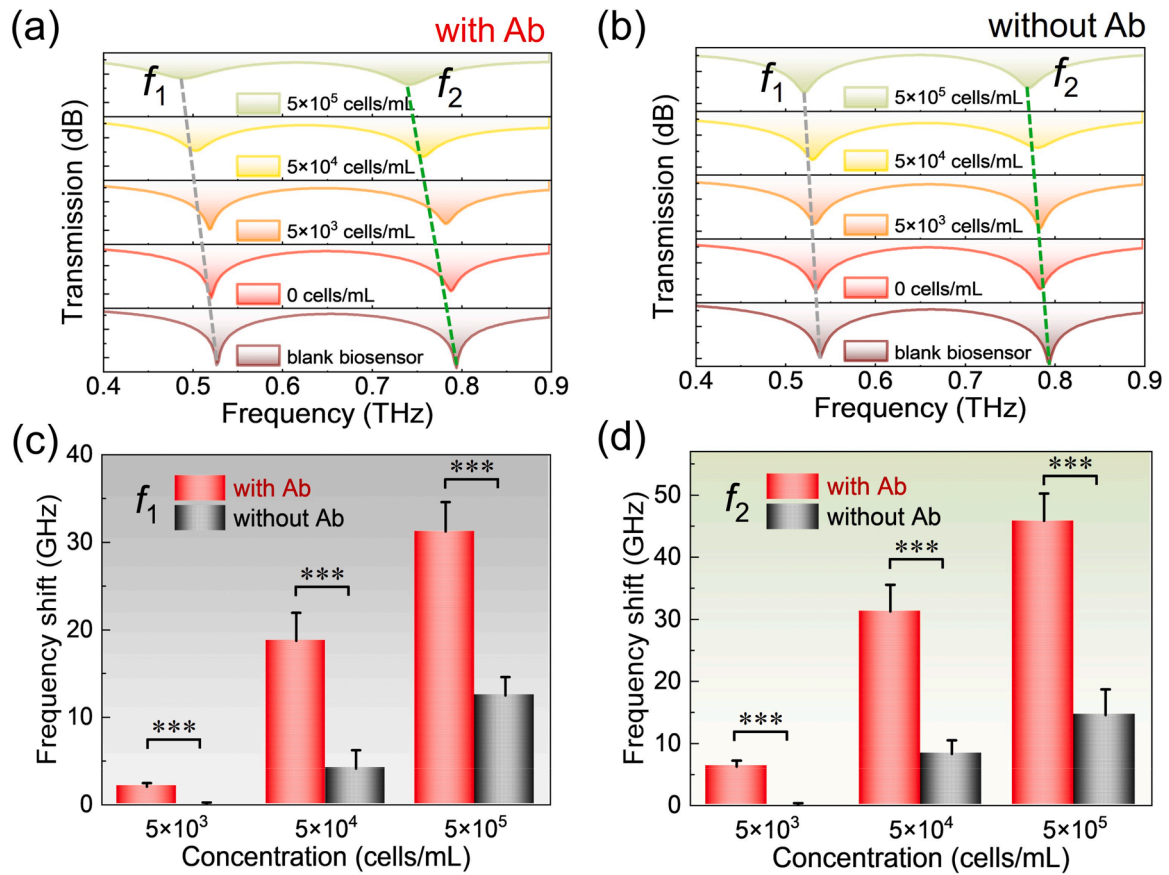


Fig. 5. The experimental transmission spectra for the live A549 cells (a) with and (b) without modification of antibody under the x-polarized. The frequency shifts at the resonance peaks (c) f_1 and (d) f_2 in the case of with and without modification of antibody, respectively. Note: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, ns: no significance. All experiments are repeated three times.

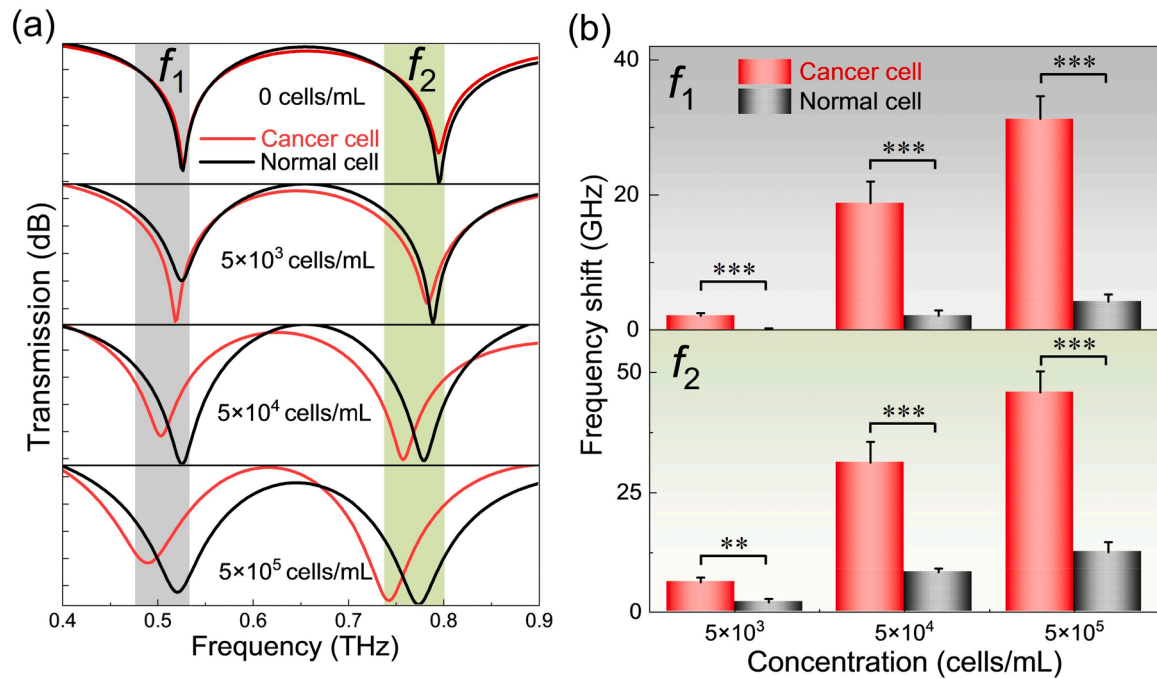


Fig. 6. (a) The experimental results for live A549 cells and normal cells with different concentrations under the x-polarized. (b) The frequency shifts for the live A549 cells and the normal cells at resonance peaks. Note: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, ns: no significance. All experiments are repeated three times.

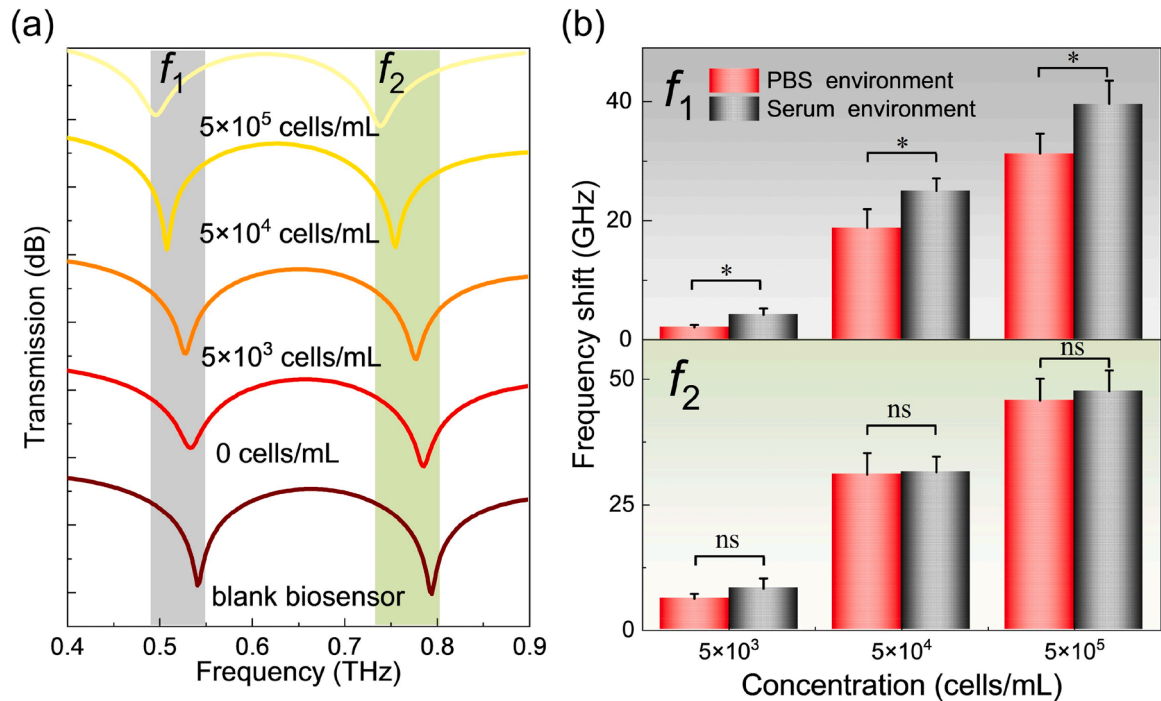


Fig. 7. (a) The experimental transmission spectra for live A549 cells with different concentrations in the serum environment under the x-polarized. (b) Comparison of the corresponding frequency shifts at two resonance peaks in the PBS and serum environments. Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns: no significance. All experiments are repeated three times.

Table 1
Comparison of this study with previous representative biosensors in the field of THz sensing.

Metasurface type	Working frequency (THz)	Cancer cell type	Cell state	LoD (cells/mL)	Specific detection	Ref.
Monolayer/EIT	0.5–3.0	Skin	Dry	1×10^5	No	[12]
Monolayer/ED	1.6–2.4	Lung/Brain	Dry	4×10^5	No	[43]
Monolayer/ED+TD	1.5–3.5	Lung	Dry	2×10^4	No	[20]
Monolayer/TD	1.0–3.0	Lung	Dry	2×10^4	No	[19]
Monolayer/EIT	1.4–3.0	Glioma	Dry	2×10^5	No	[44]
Monolayer/Fano	0.5–4.5	Lung	Dry	1×10^4	No	[18]
Monolayer/EIT	0.5–2.5	Lung	Dry	5×10^4	No	[45]
Monolayer/EIT	0.5–2.5	Oral	Dry	1×10^5	No	[46]
Bilayer/ED	0.4–0.9	Lung	Liquid/ live	1×10^3	Yes	Our

Note: LoD: limit of detection, EIT: Electromagnetically induced transparency, ED: electric dipole resonance, TD: Toroidal dipole resonance, Fano: Fano resonance
^a The expression in the original text is 200 cells/mm². For ease of comparison, it is converted to 2×10^4 cells/mL based on a sample holder of 1 cm².

Fan: Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Mengfan Han:** Writing – review & editing, Supervision. **Yifeng Li:** Writing – review & editing, Supervision. **Meng Meng:** Writing – review & editing, Supervision. **Yan Ma:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

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Data Availability

Data will be made available on request.

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